

MODULE 1 : Introduction to robotics

Introduction to robotics – Degrees of freedom, Robot types- Manipulators- Anatomy of a robotic manipulator-links, joints, actuators, sensors, controllers. Robot configurations-PPP, RPP, RRP, RRR. Mobile robots- wheeled, legged, aerial robots, underwater robots, surface water robots . Dynamic characteristics- speed of motion, load carrying capacity & speed of response. Introduction to End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and Passive grippers. Ethics in robotics - 3 laws - applications of robots.

DEFINITION

There is no clear definition of a “robot”!

The Robot Institute of America (1969) defines robot as

“.... a re-programmable, multi-functional manipulator designed to move materials, parts, tools or specialized devices through various programmed motions for the performance of a variety of tasks”.

Currently the term “robots” are used more broadly as an

“intelligent agent, physical or virtual, capable of doing a task autonomously or with guidance”.

Webster’s defined robot as

“an automatic device that performs functions ordinarily ascribed to human beings”.

ROBOT:

The term „robot“ was derived from the English translation of a fantasy play written in Czechoslovakia around 1920. „Robota“ means either a slave or mechanical item that would help its master. A robot carries out the task done by a human being. A robot may do assembly work where some sort of intelligence or decision making capability is expected.

ROBOTICS:

Robotics is the science of designing and building robots suitable for real-life applications in automated manufacturing and non manufacturing environment.

INTRODUCTION

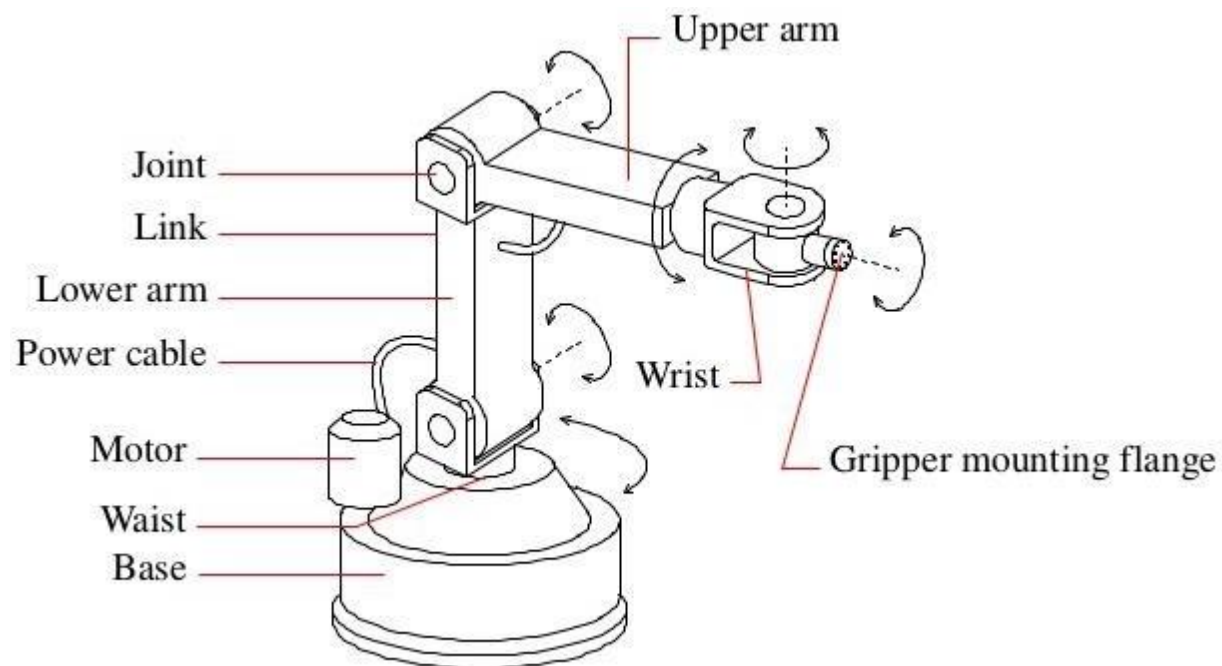
- Initial robot usage was primarily in industrial application such as part/material handling, welding and painting and few in handling of hazardous material.
- Most initial robots operated in *teach-playback* mode, and replaced „repetitive“ and „back-breaking“ tasks.
- Growth and usage of robots slowed significantly in late 1980“s and early 1990“s due to “lack of intelligence” and “ability to adapt” to changing environment – Robots were essentially blind, deaf and dumb!
- Last 15 years or so, sophisticated sensors and programming allow robots to act much more *intelligently, autonomously* and *react to changes in environments faster*.
- Present-day robots Used in:
 - *cluttered* workspaces in homes and factories,
 - Interact *safely* with humans in close proximity,
 - Operate *autonomously* in hazardous environments,
 - Used in entertainment and in *improving* quality of life.

Types of Robots

- Manipulators (Robotic Arm)
- Mobile Robots
- Aerial Robots

Manipulators (Robotic Arm)

A robotic arm is a type of mechanical arm, usually programmable, with similar functions to a human arm; the arm may be the sum total of the mechanism or may be part of a more complex robot. The links of such a manipulator are connected by joints allowing either rotational motion (such as in an articulated robot) or translational (linear) displacement. The links of the manipulator can be considered to form a kinematic chain. The terminus of the kinematic chain of the manipulator is called the end effector and it is analogous to the human hand.



Mobile Robots

Mobile robots have the capability to move around in their environment and are not fixed to one physical location. Mobile robots can be "autonomous" (AMR - autonomous mobile robot) which means they are capable of navigating an uncontrolled environment without the need for physical or electro-mechanical guidance devices. Alternatively, mobile robots can rely on guidance devices that allow them to travel a pre-defined navigation route in relatively controlled space (AGV - autonomous guided vehicle).

Legged robots are a type of mobile robot, which use articulated limbs, such as leg mechanisms, to provide locomotion. They are more versatile than wheeled robots and can traverse many different terrains, though these advantages require increased complexity and power consumption. Legged robots often imitate legged animals, such as humans or insects, in an example of biomimicry.



Legged Robot

Wheeled Robot

Wheeled robots are robots that navigate around the ground using motorized wheels to propel themselves. This design is simpler than using treads or legs and by using wheels they are easier to design, build, and program for movement in flat, not-so-rugged terrain. They are also more well controlled than other types of robots. Disadvantages of wheeled robots are that they can not navigate well over obstacles, such as rocky terrain, sharp declines, or areas with low friction. Wheeled robots are most popular among the consumer market, their differential steering provides low cost and simplicity. Robots can have any number of wheels, but three wheels are sufficient for static and dynamic balance. Additional wheels can add to balance; however, additional mechanisms will be required to keep all the wheels in the ground, when the terrain is not flat.

Aerial Robots

It's also called as Unmanned Aerial Vehicle (UAV). An **unmanned aerial vehicle (UAV)** (or **drone**) is an aircraft without a human pilot on board and a type of unmanned vehicle. UAVs are a component of an unmanned aircraft system (UAS); which include a UAV, a ground-based controller, and a system of communications between the two. The flight of UAVs may operate with various degrees of autonomy: either under remote control by a human operator, autonomously by onboard computers or piloted by an autonomous robot.

Robot Anatomy

A robot as a system, consists of a manipulator or rover, a wrist, an end-effector, actuators, sensors, controllers, processors, and software.

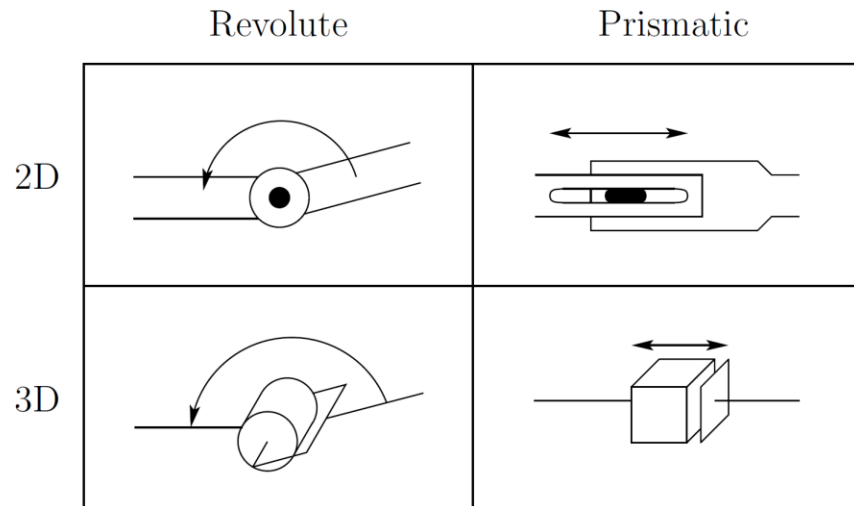
Robot Manipulators are composed of links connected by joints into a kinematic chain.

Link

A robot arm or a robot link is a rigid member that may have relative motion with respect to all other links.

Joint

Two links are connected by contact at a joint where their relative motion can be expressed by a single coordinate. Joints are typically revolute (rotary) or prismatic (translatory).

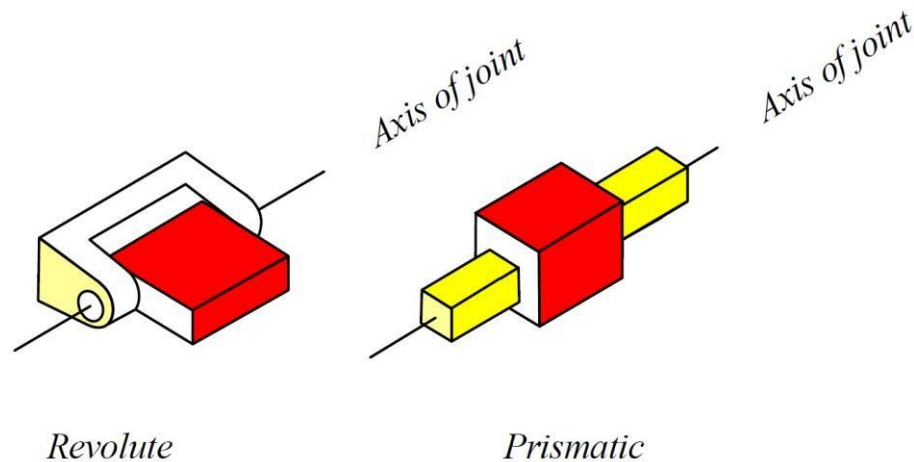


A *revolute joint (R)*, is like a hinge and allows relative rotation between two links.

A *prismatic joint (P)*, allows a translation of relative motion between two links.

Relative rotation of connected links by a revolute joint occurs about a line called *axis of joint*. Also, translation of two connected links by a prismatic joint occurs along a line also called *axis of joint*.

The value of the single coordinate describing the relative position of two connected links at a joint is called joint coordinate or joint variable. It is an angle for a revolute joint, and a distance for a prismatic joint.



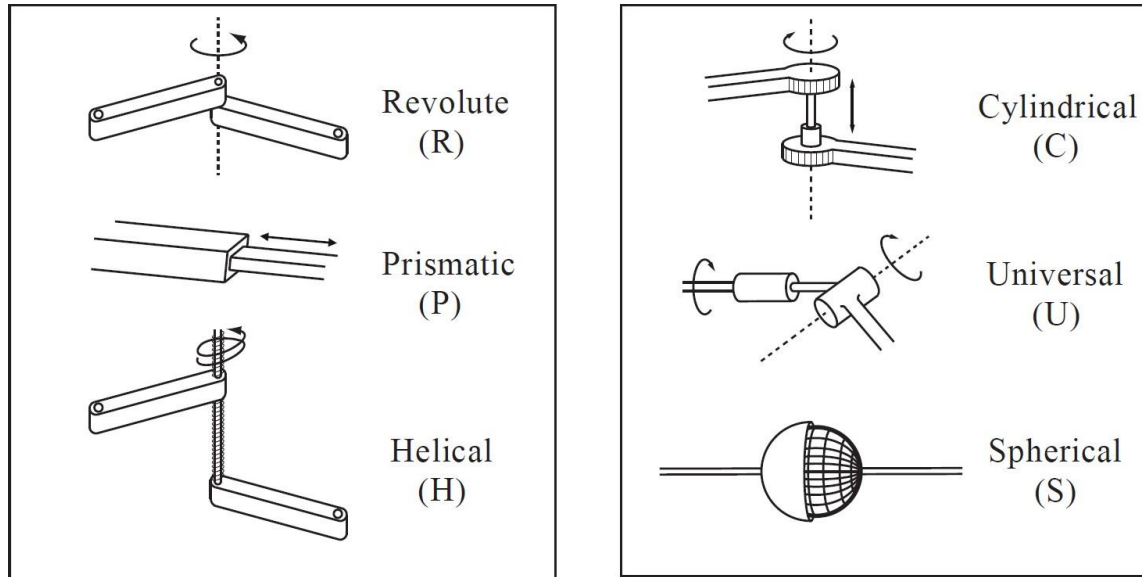
active joint

The coordinate of an active joint is controlled by an actuator. Active joints are usually prismatic or revolute

passive joint

A passive joint does not have any actuator. Passive joints are also called inactive or free joints.

There are six different lower pair joints: revolute, prismatic, cylindrical, screw, spherical, and planar. Revolute and prismatic joints are the most common joints that are utilized in serial robotic manipulators.



Degrees of Freedom and Workspace

The number of joints determines the degrees-of-freedom (DOF) of the manipulator. Typically, a manipulator should possess at least six independent DOF: three for positioning (x,y,z) and three for orientation (roll, pitch, yaw). If the robot has less than six DOF the arm cannot reach every point in its work environment. Certain applications such as reaching around or behind obstacles require more than six DOF. The difficulty of controlling a manipulator increases rapidly with the number of links. A manipulator having more than six links is referred to as a *kinematically redundant manipulator*.

workspace

The workspace of a manipulator is the total volume swept out by the end-effector as the

manipulator executes all possible motions. The workspace is limited by the geometry of the manipulator as well as mechanical constraints on the joints. For example, a revolute joint may be limited to less than a full 360 degrees of motion.

The workspace is further divided into a *reachable workspace* and a *dextrous workspace*.

- The reachable workspace is the entire set of points reachable by the manipulator.
- The dextrous workspace consists of those points that the manipulator can reach with an arbitrary orientation of the end-effector.

Obviously, the dextrous workspace is a subset of the reachable workspace.

Manipulator

The main body of a robot consisting of the links, joints, and other structural elements, is called the manipulator. A manipulator becomes a robot when the wrist and gripper are attached, and the control system is implemented. However, in literature robots and manipulators are utilized equivalently and both refer to robots.

Wrist

The joints in the kinematic chain of a robot between the forearm and end effector are referred to as the wrist. It is common to design manipulators with spherical wrists. The spherical wrist greatly simplifies the kinematic analysis effectively. This allows to have 3 joints for positioning and a spherical wrist for orientation of end-effector. Thus the arm will have 6 DoF (3 for positioning

and 3 for orientation) . We may design a wrist having one, two, or three DOF depending on the application.

end-effector

The end-effector is the part mounted on the last link to do the required job of the robot. The simplest end-effector is a gripper, which is usually capable of only two actions: opening and closing. The arm and wrist assemblies of a robot are used primarily for positioning the end-effector and any tool it may carry. It is the end-effector or tool that actually performs the work. Thus, a robot is composed of a manipulator or mainframe and a wrist plus a tool. The wrist and end-effector assembly is also called a hand.

Actuators

Actuators are drivers that move the joints in robots to change their configuration. The actuators provide power to act on the mechanical structure against gravity, inertia, and other external forces to modify the geometric location of the robot. The actuators can be of electric, hydraulic, or pneumatic type and have to be controllable.

Sensors

The sensors are elements used for detecting and collecting information about internal conditions of robot and its surrounding environmental states. Generally, joint position, velocity, acceleration, and force are the most important information to be sensed. Sensors, integrated into the robot, send information about each link and joint to the control unit, and the control unit determines the configuration of the robot.

Controller

The controller or control unit includes the processor and software. It has three roles.

1. Information role, which consists of collecting and processing the information provided by the robot's sensors.
2. Decision role, which consists of planning the geometric motion of the robot structure.
3. Communication role, which consists of organizing the information between the robot and its environment.

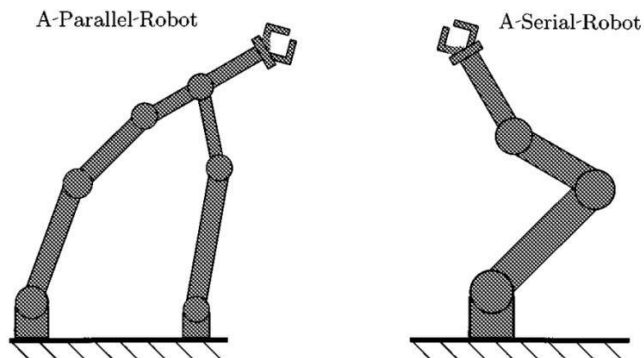
Open Kinematic chain vs Closed Kinematic chain

kinematic chain is an assembly of rigid links connected by joints to provide desired motion. This assembly is represented by a mathematical model (Kinetic model) for a robotic system. Mathematical models of the connections, or joints, between two links are termed kinematic pairs. A kinematic diagram is a schematic of the mechanical system that shows the kinematic chain. Many of the kinematic chains that are popular today have the links connected sequentially starting from a fixed base. Typically, each link is connected to the previous link in the link chain by a joint (revolute, prismatic, or spherical) and all of the joints are actuated.

In an open link chain or “*open kinematic chain*” the last link in the chain is connected from one end to a previous link but is free from the other end resulting In the robotics literature, we usually refer to open kinematic chains as “*serial robots.*”

Some kinematic chains differ from this structure in which the links are connected in series as well as in parallel combinations forming one or more closed-link loops. We will refer to these mechanisms as “*closed kinematic chains*” or “closed-chain mechanisms.” Typically, not all joints of closed-chain mechanisms are actuated. In the robotics literature, we refer to these mechanisms as “*parallel robots*.”

Figure below shows planar examples of an open kinematic chain (serial robot) and a closed kinematic chain (parallel robot). For closed-chain mechanisms, the actuators are typically placed lower (closer to the base or on the base itself) in the link chain. This makes the moving parts lighter which can lead to greater efficiency and faster acceleration at the end effector. Closed-chain mechanisms offer also greater rigidity to weight ratio. Therefore, they are more suitable for fast assembly lines.



Robot Specifications (Robot Consideration for an Application)

The various specifications of robot are :

- 1) *Work envelope*: Its the envelope or space within which the robot can manipulate the end of the wrist. It depends on the number of joints, physical size of the joints and links and the ranges of various joints.
- 2) *Load carrying capacity*: The load that can be carried by the robot depends on its physical size, construction and its capability to transmit force and torque to the end effector in the wrist. Generally for light robots this comes to around 1.5 kg. For heavier class robots this may go up to 1000Kg.
- 3) *Speed of movement*: It is the speed at which a robot is capable of manipulating its end-effector. Its governed by :
 - a) Distance to be moved;
 - b) The weight of the part to be moved;
 - c) The accuracy required in placement of the part in position.
- 4) *Repeatability*: It is the measure of the robots ability to position an object at a previously taught point in the work envelope.
- 5) *Control resolution*: It relates to the systems capability to divide the range of the total movement into closely spaced points that can be

identified. Thus , it would represent the minimum noticeable movement achievable.

- 6) *Spatial resolution*: Whereas control resolution only concerns one joint or one link, spatial resolution combines the control resolution of all motions and also considers the mechanical errors in the points and associated links.
- 7) *Mechanical errors*: The mechanical errors result from; backlash in gears, hysteresis, deflection of links, hydraulic leaks etc.
- 8) *Accuracy*: It is the measure of the ability of robot to position the end of the wrist at the desired location in the work envelope.
- 9) *Stability*: It relates to the amount of overshoot and oscillations in robot motion as it is about to reach a certain location.

Accuracy , Repeatability and precision

The **accuracy** of a manipulator is a measure of how close the manipulator can come to a given point within its workspace.

Repeatability is a measure of how close a manipulator can return to a previously taught point.

Most present day manipulators are highly repeatable but not very accurate. The reason is that the angular/translator position of the joints have slight error. The primary method of sensing this positioning errors in most cases is with position encoders located at the joints. There is

typically no direct measurement of the end-effector position and orientation. One must rely on the assumed geometry of the manipulator and its rigidity to infer (i.e., to calculate) the end-effector position from the measured joint angles.

Thus, accuracy is affected by:

- computational errors,
- machining accuracy in the construction of the manipulator,
- flexibility effects such as the bending of the links under gravitational and other loads
- gear backlash
- other static and dynamic effects.

It is primarily for this reason that robots are designed with extremely high rigidity. Without high rigidity, accuracy can only be improved by some sort of direct sensing of the end-effector position, such as with vision.

The computer generally stores position error in accuracy and is used in the programming to take care that error doesn't add up in the long run. Repeatability therefore is affected primarily by the **controller resolution or precision**. Controller resolution means the smallest increment of motion that the controller can sense. prismatic joints, typically have higher resolution than revolute joints, since the straight line distance traversed by the tip of a linear axis between two points is

less than the corresponding arc length traced by the tip of a rotational link.

Classification of Robot Based on Control Systems

1. Point-to-point (PTP) control robot: is capable of moving from one point to another point. The locations are recorded in the control memory. PTP robots do not control the path to get from one point to the next point. Common applications include component insertion, spot welding, hole drilling, machine loading and unloading, and crude assembly operations.
2. Continuous-path (CP) control robot: with CP control, the robot can stop at any specified point along the controlled path. All the points along the path must be stored explicitly in the robot's control memory. Typical applications include spray painting, finishing, gluing, and arc welding operations.
3. Controlled-path robot: the control equipment can generate paths of different geometry such as straight lines, circles, and interpolated curves with a high degree of accuracy. All controlled-path robots have a servo capability to correct their path

ROBOTS IN INDUSTRY

Introduction

A robot is a reprogrammable, multifunctional machine designed to manipulate materials, parts, tools, or specialized devices, through variable programmed motions for the performance of a variety of task. It can conveniently be used for variety of industrial tasks. Today 90% of all robots used are found in factories and they are referred to as industrial robots. Robots are slowly finding their way into warehouses, laboratories, research and exploration sites, power plants, hospitals, undersea, and even outer space.

Few of the advantages for which the robots are attractive in industrial uses are as follows :

- ✓ Robots never get sick or need to rest, so they can work 24 hours a day, 7 days a week.
- ✓ When the task is dangerous for a person, Robots can do the work instead.
- ✓ Robots do not get bored. So the work that is repetitive is of no problem for a robot.

INDUSTRIAL APPLICATIONS

There are certain industrial tasks robots do very well like material transfer, machine loading, spot welding, continuous arc welding, spray coating, material removal, cutting operations, assembling operations, part inspection, part sorting, part cleaning, part polishing, and a dozen more specialized tasks. In this section, some of these applications will be elaborated.

Material Handling : The time spent to transfer a workpiece from one station to the next is high in industries. Up to 95% of the time involved in manufacturing a part is composed of transfer and waiting time, and only about 5 percent of the total time is the part in the actual processing time. Loading and unloading machine tools are the major applications of robots. Robots are utilised to load and unload machine tools in two basic configurations :

- (a) a robot tending a single machine, and
- (b) a robot serving several machines.

The items being handled by robots may weigh anything from a few grams to some tonnes. Obviously the type of robot applied will also vary considerably, from lightweight non-servo-controlled pneumatic robots,

through medium sized electric types, to massive hydraulic manipulators.

For material handling applications point to point control is necessary. The controller should be able to store a large number of points and have the facility to easily program horizontal and vertical increments.

Welding : The welding operations performed by robots are thermal processes in which the metals are joined by melting or fusing their contacting surfaces. These processes can be grouped under two classes,

- a) Where no filler material is added to the joint interface,
- b) In which a filler material of the same type as the parent metal is added.

Accordingly, there are two types of welding operations performed by the robots :

- a) spot welding.

Spot welding falls in the first category, in which the two pieces of metal are joined only at certain points. The required heat is generated by the passage of an electric current through the metals at the point where they are to be joined. Spot welding is frequently used in the automotive industry to join

thin sheet metals.

b) arc welding.

Arc welding falls in the second category in which the two metals are joined along a continuous path. The required heat is provided by an electric arc generated between the electrode and the metals. Arc welding is needed, for example, in sealing a container against leakage.

Each of these welding operations requires a different type of equipment and a different control system for the robot arm.

Spray Painting

: The unhealthy and unpleasant environment of the painting booth in industry are:

- The solvent materials that are used in spray painting are toxic.
- The painting area must be dust-free and temperature-controlled.
- The noise arising from the air discharge through the painting nozzles can cause irreversible damage to the ears.

For all these reasons, spray painting became one of the first applications of robots. The requirement for robots in spray painting are different from those of other robot

applications, and therefore many robot manufacturers offer a robot dedicated to this one application.

The spray painting robots are of Continuous Path capability and have the following characteristics:

- (a) high level of manipulator dexterity
- (b) large working volume for small-base manipulator
- (c) compact writs
- (d) small payload
- (e) low accuracy and repeatability.

The requirements for repeatability and resolution are the least severe in painting robots. The exact location of end points is not critical, and in many jobs can be even outside the painted surface.

Assembling : Assembling with industrial robots is mainly used for small products such as electrical switches and small motors. Robotised assembly systems are programmable and therefore provided a cost-effective solution for the assembly of small batch sizes and for batches containing different products. Tactile or optical sensors may be added to the assembly robot to tackle more complex assembly tasks.

Some assembly tasks require the participation of more

than one robot. In order to reduce the cost per arm, there are systems in which several cartesian arms can use the same base and share the same controller. Assembly robots can be designed in any coordinate system, cartesian, cylindrical, spherical, or articulated. However, many tasks require only vertical assembly motions, such as the assembly of printed circuit boards. For these applications a class of robot known as the *selective-compliance-assembly robot-arm* (SCARA) type robot, and was developed in Japan.

Machining : There are five basic types of machine tools to perform machining : drilling machine, lathe or turning machine, milling machine, shaper, and grinder. Out of all these machining operations, only drilling is being successfully done with robots, and mainly in the aircraft industry. Another application related to machining which is performed by robots is deburring metal parts. Most metal parts made by machining operations (either by machine tools or by mass production machines) contain burrs, that is, rough edges or ridges left on the machined surfaces. The removal of these burrs can be done by robots.

State-of-the-art Applications of Robots

Present day robots are also used in medical surgery, space explorations, and others, which are explained next.

Medical : Medical robots have found applications mainly in surgery. It is a highly interactive process and many surgical decisions are made in the operating room. The goal of surgical robots is not to replace the surgeon with a robot, but to provide the surgeon with a new set of very versatile tools that extend his or her ability to treat patients. Hence, medical robotic systems are surgical assistants that work cooperatively with surgeons. A special subclass of these systems is often used for remote surgery. Currently, there are two main varieties of surgical assistant robot are available.

Surgeon Extender

It is a robot controlled directly by the surgeons. They augment or supplement the surgeon's ability to manipulate surgical instruments during surgery. The promise of these systems is that they can give even average surgeons superhuman capabilities such as elimination of hand tremor or ability to

perform dexterous operations inside the patient's body, etc. As a result, casualty rates are reduced, and operative times are shortened

Auxiliary Surgical Support

It generally works side-by-side with the surgeon and perform such functions as endoscope holding or retraction. These systems typically provide one or more direct control interfaces such as joysticks, head trackers, voice control, or the like. However, the surgeon should be attentive during their use, e.g., while using robot vision to keep the endoscope aimed at an anatomic target.

Surgical robots must be compatible with the operating theatre. The robot must have sufficient strength, accuracy, and dexterity for its intended use. It must be placed where it can work on the patient while also allowing access by clinical staff. Usually, this is done by mounting the robot to the operating table or placing it on the floor beside the patient.

Mining : In order to enhance productivity, access unworkable mineral seams, and reduce human exposure to the inhospitable environment of dust, noise, gas, water,

moving equipment and robots are used. In coal mining room-and-pillar mining is accomplished by repetition of a well-defined cycle of cutting coal, removing coal, and supporting the roof. When a robot is used the ability to maneuver it in highly constrained corridor-like environments, and to accurately cutting coal is required. Data from the laser range scanner can be used to model the environment and to compute the position of the robot.

- Space** : Explorations of the planets, moons and other near bodies in space are a clear goal for the international space science community. There are a variety of tasks that the robots can do in space including
- space manipulation (servicing equipment in space)
 - surface mobility (planetary exploration)
 - Robotic colonies (outposts that are either self-sustaining, or preparatory for human colonies).
 - In addition, robots may perform scientific experiments that include sample and return of planetary atmosphere or terrain, manipulating the environment (moving rocks, drilling, etc.), testing the composition of the atmosphere and other tests using arbitrary scientific equipment.

There are three characteristics that a robot should have for space missions :

- c) Compactness and Lightness: The cost of sending robot into space is directly correlated to its size and weight.
- d) Robustness: Missions often have only one attempt to succeed (usually at great cost).
- e) Versatility and Adaptability: In exploration where the environments are inherently unknown, adaptability must be high to increase the chance of success.

Underwater : Underwater applications of robots involve prospecting for minerals on the floor of the ocean (e.g., nodules of manganese), salvaging of sunken vessels, and the repair of ships either at sea or in dry dock. In the latter case, a prototype version of a mobile robot that is used to clean barnacles from the sides of ships has been built and tested in France by a ship-building company. This rather remarkable tripod is capable of moving in either air (i.e., above the waterline) or in water. It grips the ships sides with both vacuum and magnetic feet, a technique that has proven to be reliable. The scrubbing action is produced by a rotating brush mounted on the end of a rotary axis arm.

Defence : The defence people, namely, the air force and navy, are both interested in mobile fire fighters. These devices would be equipped with infrared sensors and could react more quickly than people in an emergency and in extremely hazardous situations.

Other defence applications of robots will be on the battlefield itself. Although it is not inconceivable that robots might someday be used to fight other robots, more realistic short-term applications from the defence's point of view would be in the areas of surveillance (e.g., guard and sentry duty), mine sweeping, and artillery-loading devices. The application of robots for surveillance and guard duty can be in power generating plants, oil refineries, and other large civilian facilities that are potential targets of the terrorist groups. The robots for these applications would probably be mobile (running on wheels, treads, or tracks), equipped with some form of vision system and other types of sensors (e.g., infrared), and even have defensive and/or offensive capability.